

Problem 6.14

An investor puts 100 into Fund X and 100 into Fund Y . Fund Y earns compound interest at the annual rate of $j > 0$, and Fund X earns simple interest at the annual rate of $1.05j$. At the end of 2 years, the amount in Fund Y is equal to the amount in Fund X . Calculate the amount in Fund Y at the end of 5 years.

The amount in Fund X after 2 years is

$$100(1 + 1.05j(2)) = 100(1 + 2.1j),$$

since Fund X earns simple interest.

The amount in Fund Y after 2 years is

$$100(1 + j)^2,$$

since Fund Y earns compound interest.

We are told these amounts are equal, so

$$100(1 + 2.1j) = 100(1 + j)^2.$$

Dividing by 100,

$$1 + 2.1j = (1 + j)^2 = 1 + 2j + j^2.$$

Thus,

$$j^2 - 0.1j = 0,$$

so

$$j(j - 0.1) = 0.$$

Since $j > 0$, we must have

$$j = 0.1.$$

Therefore, the amount in Fund Y at the end of 5 years is

$$100(1 + j)^5 = 100(1.1)^5 = 100(1.61051).$$

Hence,

$$\boxed{161.051}.$$